

Name: <u>Solution</u>	ID:	Section: <u>0506</u>
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Consider a binary number, $W = 0.000111010 \times 2^8$. Convert W in 30-bit IEEE-754 format where the size of the fraction field is 18 bits.

(1)	(11)	(18)
sign bit	exponent	fraction

$$\begin{aligned}
 W &= 0.000111010 \times 2^8 \\
 &= 1.11010 \times 2^{8-4} \\
 &= 1.11010 \times 2^4 \rightarrow \text{Actual Exp.}
 \end{aligned}$$

a) What is the Bias of this format?

Answer: $\text{Bias} = 2^{n-1} - 1$

$$= 2^{10} - 1 = 1024 - 1 = 1023$$

b) Find the value of the Biased exponent. (answer must be in binary) [1]

Answer: $\text{Biased Exp.} = \text{Bias} + \text{Exponent}$

$$= 1023 + 4$$

$$= 1027 = 100\ 0000\ 0011$$

c) Find the value of the fraction field. (answer must be in binary) [1]

Answer: $1101\ 0000\ 0000\ 0000\ 00$

d) What is the value of the sign field? Answer: 0 [1]

e) Finally, complete the 30-bit representation and convert your answer into Hex. [1]

Answer:

0	100 0000 0011	1101 0000 0000 0000 00
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